BARATH R

Software Developer | Full-Stack · AI-NLP Research

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PROFESSIONAL SUMMARY

Software Developer with 2.5+ years of experience at AU-KBC Research Centre, Anna University, delivering across full-stack web development, AI/NLP integrations, and Android application development. Hands-on across the entire stack from React and Spring Boot on the frontend/backend, to building and deploying NLP pipelines, chatbots, and annotation tools, to native Android apps with Java. Comfortable owning projects end-to-end in fast-moving, research-driven environments.

PROFESSIONAL EXPERIENCE

Software Developer | *AU-KBC Research Centre, Anna University - Chennai,* February 2024 – Present

Discourse Platform - *React · Spring Boot · WebSocket · SQL* | Led 100+ Researchers

- Led and architected a large-scale, on-premise NLP research platform with two core layers - Annotation and Model Development - built entirely on WebSocket-driven real-time communication with minimal REST API dependency, ensuring low-latency collaboration across 100+ researchers.
- Annotation Layer: Designed a dual-mode annotation system supporting Manual Annotation (POS, Morphological, Chunking, NER tagging) and Automatic Annotation using in-house CRF models; integrated a Corpus Management module with file management capabilities for structured dataset handling and multi-format corpus storage.
- Multi-tenant Architecture: Engineered role-based access supporting both standalone individual users and separate organisations on the same platform, with well-structured SQL schemas storing only local file paths - all data, datasets, and models stored entirely on-premise with zero cloud dependency.
- Model Development Layer: Built an end-to-end model lifecycle module allowing researchers to develop, train, and run inference on deep learning models directly within the platform, bridging annotation output to model training pipelines seamlessly.

LCAuthoring Interface - *HTML · CSS · JavaScript · Java · MongoDB* | Co-developed

- Built a web-based multilingual text annotation tool supporting multiple Indian regional languages, enabling researchers to annotate plain text and CSV format datasets through an intuitive interface.
- Designed the full stack - HTML/CSS/JS frontend, Java backend, and MongoDB for scalable annotation storage - to handle large-scale linguistic data workflows at the research centre.

Anna University Chatbot - *CRF Models · Rule-based NLP · Java* | Co-developed

- Co-built an NLP-powered chatbot for Anna University capable of answering student queries on admissions, fee structure, and academic details.
- Integrated Anaphora Resolution logic to correctly interpret contextual references in multi-turn conversations, significantly improving response accuracy over standard rule-based systems.
- Combined CRF model predictions with rule-based logic for a hybrid approach to intent classification and response generation.

Bhashayatra - AI Translation & Transcription App - *Java · Native Android · In-house NLP Models* | Live on Google Play Store

- Co-developed and deployed a production Android application supporting real-time translation and transcription across multiple Indian and regional languages.
- Integrated in-house AU-KBC translation and transcription models into a consumer-facing mobile product, bridging research-grade NLP into real-world usage via Play Store deployment.

Research Data Pipeline - *Python · Web Scraping*

- Built web scraping pipelines using Python to extract and process data from diverse online sources, supporting research data collection and corpus building for NLP model training.

PROJECTS

VisionX - Medical X-Ray Display & Enhancement App - *Java · Android TV · OpenCV · ImageProcessing*
Client Project

- Sole developer of a native Android TV application for a medical client, enabling real-time X-ray image capture, enhancement, and display directly from X-ray machine hardware via cable input.
- Implemented image enhancement algorithms using OpenCV and custom mathematical logic to remove salt-and-pepper noise and improve diagnostic image clarity on large TV displays.
- Handled the full pipeline - raw signal capture from hardware to image processing to enhanced real-time display - entirely independently for a medical domain client.

TECHNICAL SKILLS

Languages: Java, JavaScript (ES6+), Python, SQL, HTML5, CSS3

Frontend: React.js, Tailwind CSS, Three.js, Responsive Design, Component-driven Architecture

Backend: Spring Boot, Express.js, REST APIs, WebSockets

Mobile: Native Android Development (Java), Android TV, Android SDK, Google Play Store Deployment

AI / ML & NLP: Machine Learning, NLP Pipelines, CRF Models, Model Training & Fine-tuning, Chatbot Development (Rule-based & AI-powered), Data Annotation, Web Scraping (Python), OpenCV, Image Processing

Databases: MySQL, MongoDB

Tools & DevOps: Git, GitHub, Docker, Firebase

EDUCATION

Bachelor of Engineering, Computer Science and Engineering

Jeppiaar Engineering College, Chennai - Anna University | CGPA: 8.3 / 10 | April 2023

Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Web Technologies, Operating Systems, Object-Oriented Programming, Software Engineering, Computer Networks

HONORS & AWARDS

- **Naan Mudhalvan Achiever -** Awarded by the Chief Minister of Tamil Nadu for outstanding technical achievement.
- **1st Place, INSTAMIND Hackathon** - Solved a real-world logistics problem for DUNZO under competitive conditions.
- **3rd Place, National Hackathon - JNN Institute** - Built a Waste Management Segregation system at national level.
- **IBM SWMS IoT Project Recognition** - Recognised by IBM for contributions to a Smart Waste Management System IoT project.